

iADD: Improving Alignment and Diversity in Diffusion Policy Optimization

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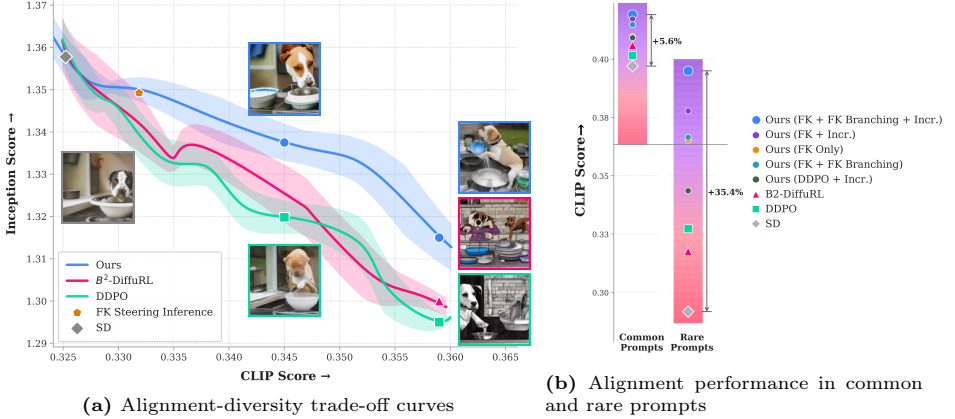


Fig. 1: Performance in alignment and diversity. (a) Our method, iADD, achieves a higher CLIP Score at any given diversity level compared to other methods. Notably, as reward increases, competing methods (DDPO and B^2 -DiffuRL) produce increasingly cartoonish, over-saturated outputs, whereas our method preserves photorealistic quality throughout training. The visualized images are for the prompt “a dog washing dishes”. (b) Alignment performance on common and rare prompts, evaluated at constant Inception Score(IS) for each method. Rare prompts represent conditions that inherently yield low-reward outcomes under the pretrained model.

Abstract. Reinforcement learning based post training of diffusion models, such as Denoising Diffusion Policy Optimization (DDPO), optimizes a reverse diffusion process under a reward function. However, current approaches to reward optimizations do so at the cost of diversity and quality. In this paper, we provide better tradeoffs through careful theoretical considerations and method design. We analyze the theoretical framework and mathematically demonstrate that *only-latter timestep* updates of diffusion model may be harmful for diversity contrary to the conclusions presented in a previous work. Additionally, we propose an incremental Feynman-Kac training based on strong theoretical foundations in order to achieve the best-yet alignment-diversity tradeoffs. We perform extensive experiments and compare our method against related diffusion policy optimization approaches in three different tasks and also provide strong

ablations for each component, thus validating strong performance gains in both alignment and diversity.

Keywords: Reinforcement Learning · Diffusion Policy Optimization

1 Introduction

Diffusion Models are currently the gold standard for generation and editing tasks on images [19, 32, 37, 45, 46, 54–56], videos [5, 18, 20] as well as 3D scenes [14, 22, 57] or objects. Through time-dependent iterative movement defining a path, diffusion methods learn to reach the target data distribution, exhibiting strong generalization and diversity [11, 30, 37]. Nevertheless, the quality of the modeling is limited by the target distribution samples; thus in most cases generation quality is naturally impacted for rare classes and specific user constraints. Reinforcement Learning (RL) [38, 47, 48, 59] has the potential to address the problem, without using additional data, in the ideals of learning solely through experience [49].

Recent works [4, 12, 24, 42, 60, 62, 64] have precisely addressed the problem of finetuning the reverse diffusion process online, by applying the RL algorithm [59]. However, naive policy optimization of diffusion models can lead to mode collapse and general quality degradation of the generated samples [12, 24]. This is reflected in the *alignment-diversity tradeoffs*, see Fig. 1.

Circumventing mode collapse and general quality degradation in lack of additional preference data [27, 29, 58, 63], though important is highly challenging. Thus, most approaches tend to be domain specific, and thus are validated on specific tasks. A more generic and theoretically grounded study of the alignment-diversity tradeoffs under the constraints of additional preference data-free rewards, remains under-explored. Existing solutions often perform regularization, *e.g.* through selected timestep updates [2, 24] or via measuring the KL divergence [12]. Specifically, [24] intuitively argues that reward backpropagation is more reliable for latter timesteps and thus do not update the model on earlier timesteps. On the other hand, [2, 34] refine middle timesteps based on the observation of how semantics are constructed in diffusion models [6, 11]. Similarly, [12, 64] specifically consider the Denoising Diffusion Probabilistic Model (DDPM) in order to propose reward optimization strategies. Furthermore, [64] proposes a continuous time formulation in DDPM for RL policy optimization.

In this paper, we study the alignment-diversity tradeoff in finetuning discrete-time denoising diffusion model through early intervention as well as careful regularization. We formulate the tradeoff problem as that of attaining the minimum change in the target samples required for the alignment, while preserving the original diversity through structural regularization, and target distribution’s volume preservation. We discover that attaining higher alignment with minimal diversity tradeoff requires early intervention via two processes. The first and retrospectively obvious solution is to update the early timesteps of the reverse denoising path, contrary to [24] that updates the late timesteps. We provide actionable theoretical results that untangles the puzzle of timestep selection.

The results justify partial timestep selection from the viewpoint of target distribution’s volume preservation. Moreover, the theoretical analyses suggest an approach, which incrementally increases the per-iteration sparse timestep updates, over the course of the finetuning. Additionally, our second solution is to perform early intervention by looking ahead in the denoising path. Specifically, we propose the discrete-time implementation of the Feynman-Kac path guidance [9, 26, 28, 50, 52, 53] during training in the following two ways: discontinuing least promising paths with low reward expectations to improve the alignment and branching the promising ones in order to encourage diversity. Table 1 shows the relative impact and relevant theory behind the method’s components.

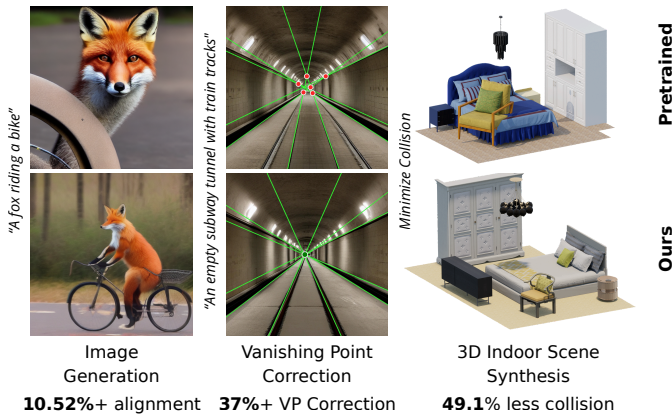


Fig. 2: Versatility across diverse generation tasks.

Our final method, termed, iADD thus employs both early timestep update, early intervention as well as structural regularization in order to obtain a balanced approach to reward-based finetuning of diffusion models. We perform experiments on three different tasks, rare prompt-based image generation, vanishing point correction in image generation and 3D scene generation, in order to show the versatility. Our comparisons show clear improvements in alignment as well as diversity/quality of generation in all three tasks compared to the state-of-the-art approaches. In summary, our main contributions are as follows:

- We provide theoretical analysis of the impact of timestep selection strategies on diversity.
- We propose iADD, a novel training method that integrates an incremental timestep curriculum with discrete-time Feynman-Kac (FK) branching to optimally balance reward alignment and diversity.
- Extensive experiments across image and 3D scene generation demonstrate that iADD is task-agnostic and achieves the best-yet alignment-diversity trade-offs, significantly outperforming existing baselines.

Table 1: Disassembly of our Method’s components and their roles. Up arrow (↑) denotes improvement and down arrow (↓) denotes worsening. Additional reference to sections or theoretical justifications are mentioned in the most relevant column.

Components		Alignment	Diversity	Rarity	Training Eff.
Timesteps	Early timesteps	↑	↑, Prop. 1	↑	↑
	Timestep sampling	↑	↑, Prop. 1, 2	↑	↑
	Incremental timesteps	↑	↑, Remark. 2	↑	↑
.....					
FK	FK Guidance	↑, Sec. 4.3	↑	↑↑	↓
	FK Branching	↑	↑↑, Sec. 4.3	↑	↓

2 Related Work

Inference-time adjustments. In order to improve alignment-diversity, reward metrics are also used solely at inference time. Such training free-approaches [51, 52, 56] utilize inference time measures to maximize the reward for specific generated samples. However, these techniques are also susceptible to degradation of sample quality at higher reward and moreover do not scale well during inference.

Preference-based optimization. RL when used with additional preference data allows for advanced techniques that can preserve diversity. Such approaches are the counterparts of RL with Human Feedback (RLHF) [7], popularly used in Large Language Models finetuning [65], in the diffusion policy optimization. Methods in the category primarily follow the Direct Policy Optimization (DPO) [44, 58] strategies either in discrete-time formulation [58] or continuous-time [25]. However, they are severely limited by the requirement of human preference data, requiring an offline strategy.

Online RL for Diffusion Models. Our work is on online RL finetuning of diffusion models [3, 4, 8, 12, 24, 61], which has seen limited applications in diffusion models, largely due to the difficulty of the joint alignment-diversity optimization. In the topic, DDPO [4] is a seminal work that finetunes the complete trajectory as discrete steps in the Markov chain. DPOK [12] proposed a KL-divergence regularization using the DDPM formulation. Notably, [24] shows the challenges of the sparse terminal rewards [1, 10, 15, 17, 23, 33, 35, 36] and suggests solely updating the later timesteps of denoising, while also incorporating contrastive pairs in branches for improving diversity. A recent work also formulates the continuous-time online RL for diffusion models [64] in the DDPM formulation while also defining each path’s value function at any given time similar to a Feynman-Kac potential. Other approaches [8, 41] use task-specific considerations for online finetuning of diffusion models. However, current approaches do not treat time-step selection under careful theoretical considerations and under-exploit the denoising paths in reward optimization leading to poorer tradeoffs.

Table 2: Summary of methods and techniques across training, inference, and diffusion formulation. Our method is designed for the DDIM formulation.

Method	Training			Inference		Formulation
	Incr.	FK	Branch.	No Test-time	Scaling	
DPOK [12]	$\times$	$\times$	$\times$		$\checkmark$	DDPM
Score-as-Action [64]	$\times$	$\checkmark$	$\times$		$\checkmark$	DDPM
DDPO [4]	$\times$	$\times$	$\times$		$\checkmark$	DDIM
B ² -DiffuRL [24]	$\times$	$\times$	$\checkmark$		$\checkmark$	DDIM
FK Steering & FK Correctors [51, 52]	$\times$	$\times$	$\times$		$\times$	DDIM
Ours	$\checkmark$	$\checkmark$	$\checkmark$		$\checkmark$	DDIM

3 Preliminaries

Discrete-Time Diffusion Models. Diffusion models [54] are a class of latent variable models that learn to approximate a target data distribution by systematically reversing a predefined Markovian corruption process. Our work specifically builds upon discrete-time formulations, where the generation trajectory unfolds over a finite set of steps $t \in \{0, \dots, T\}$.

Forward Process. The forward process progressively corrupts a data sample $x_0 \sim q(x_0)$ by injecting Gaussian noise according to a fixed variance schedule $\beta_1, \dots, \beta_T$. This trajectory is defined as a Markov chain:

$$q(x_{1:T} | x_0) = \prod_{t=1}^T q(x_t | x_{t-1}), \quad (1)$$

where the transition dynamics are given by $q(x_t | x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t} x_{t-1}, \beta_t I)$. For a sufficiently large number of steps T , the final state x_T converges to an isotropic Gaussian noise distribution $\mathcal{N}(0, I)$.

Reverse Generative Process. The core objective of a diffusion model is to learn the reverse of this trajectory, constructing clean data samples starting from pure noise x_T . The reverse process is modeled as a parameterized Markov chain, often conditioned on auxiliary context c (e.g., text prompts):

$$p_\theta(x_{0:T} | c) = p(x_T) \prod_{t=1}^T \pi_\theta(x_{t-1} | x_t, c), \quad (2)$$

where $\pi_\theta(x_{t-1} | x_t, c)$ is approximated by a deep neural network predicting the parameters of the Gaussian transition necessary to denoise x_t into x_{t-1} .

Denoising Diffusion Implicit Models (DDIM) DDIM [55] generalizes the standard DDPM [19] forward process to a family of non-Markovian diffusion processes that share the same training objective. This decouples the number of inference steps from the number of training timesteps, enabling sampling over an arbitrary subsequence $\mathcal{T}_{sub} \subset \{1, \dots, T\}$ without any retraining. Crucially, at each denoising step t , DDIM produces an explicit estimate of the clean sample x_0 via:

$$\hat{x}_0 = \frac{x_t - \sqrt{1 - \alpha_t} \epsilon_\theta^{(t)}(x_t)}{\sqrt{\alpha_t}} \quad (3)$$

which we leverage to compute reward estimates from intermediate timesteps, enabling us to guide the generative trajectory before the final sample is produced.

Denoising Diffusion Policy Optimization (DDPO) Denoising Diffusion Policy Optimization (DDPO) [4] treats the iterative reverse diffusion process as a multi-step Markov Decision Process (MDP). At each timestep t , the state is defined as $s_t \triangleq (c, t, x_t)$, the action as $a_t \triangleq x_{t-1}$, and the policy is parameterized by the reverse transition probability $\pi(a_t | s_t) \triangleq \pi_\theta(x_{t-1} | x_t, c)$.

The objective is to optimize a downstream reward $R(x_0, c)$ based on the final generated sample x_0 and the conditioning context c . Because each reverse transition is an isotropic Gaussian, exact per-step log-likelihoods are tractable. DDPO optimizes this MDP using an importance-weighted policy gradient estimator (DDPO-IS), allowing for multiple gradient updates using data from a previous policy parameters θ_{old} :

$$\nabla_\theta \mathcal{F}(\theta) = \mathbb{E} \left[\sum_{t=0}^T \frac{\pi_\theta(x_{t-1} | x_t, c)}{\pi_{\theta_{\text{old}}}(x_{t-1} | x_t, c)} \nabla_\theta \log \pi_\theta(x_{t-1} | x_t, c) \cdot R(x_0, c) \right]. \quad (4)$$

However, standard DDPO updates all timesteps equally and relies solely on a single final reward evaluated on the complete trajectory. This sparse terminal reward and uniform credit assignment leads to reward hacking, which naturally motivates our proposed particle-based local advantage estimation and incremental training scheduling.

4 Method

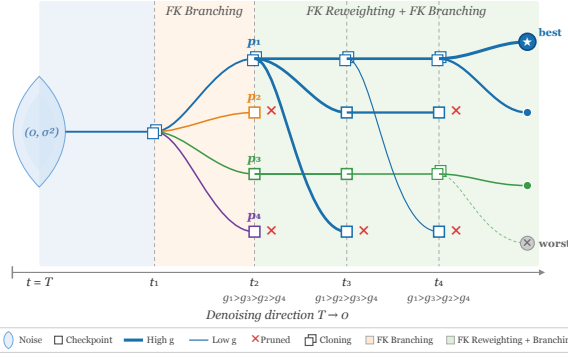
4.1 Problem formulation

Let $\pi_\phi(\mathbf{x}_{t-1} | \mathbf{x}_t, c)$ denote a pretrained conditional diffusion *reverse* transition model (with parameters ϕ) used for sampling, where c is a condition (e.g., a text prompt) and $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. Sampling produces a trajectory $\tau = (\mathbf{x}_T, \mathbf{x}_{T-1}, \dots, \mathbf{x}_0)$ by iterating $\mathbf{x}_{t-1} \sim \pi_\phi(\cdot | \mathbf{x}_t, c)$ for $t = T, \dots, 1$. This induces a marginal distribution over outputs $\mathbf{x}_0$, which we denote by $p_\phi(\mathbf{x}_0 | c)$.

Our goal is to adapt the sampler to align generated samples with user preferences while preserving the original prior and diversity. We formalize preferences via a reward function $R(\mathbf{x}_0, c) \in \mathbb{R}$ that scores the final generated sample (e.g., aesthetics, constraint satisfaction, or a task-specific metric), and seek to increase expected reward while preserving sample quality and diversity.

We parameterize the *trainable* sampler by $\pi_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t, c)$, initialized from the pretrained model ($\theta \leftarrow \phi$) and optimized using reinforcement learning (DDPO) to maximize a sparse terminal reward:

$$\mathcal{F}(\theta) = \mathbb{E}_{\tau \sim \pi_\theta} [R(\mathbf{x}_0, c)], \quad (5)$$

(A) FK Sampling ($N = 4$ particles)

(B) Incremental Timesteps

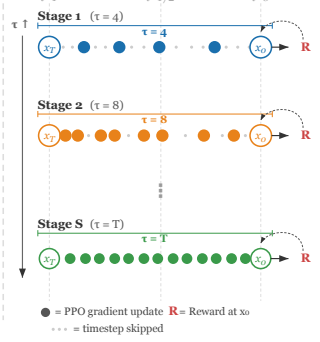


Fig. 3: Overview of FK Sampling and Incremental Timestep Training. (A) FK Sampling: Our approach employs a Feynman-Kac (FK) particle filter for sampling. Starting from Gaussian noise at $t = T$, trajectories $(p_1, \dots, p_4)$ are evaluated at discrete checkpoints t_n based on their potentials g . The *FK Branching* mechanism duplicates high-potential particles while pruning low-potential ones, effectively redirecting the stochastic denoising paths toward regions of higher expected reward. (B) Incremental timestep updates: During training, we utilize a progressive stage-wise strategy where the cardinality of the updated timesteps $\mathcal{S}$ increases.

where a trajectory $\tau = (\mathbf{x}_T, \mathbf{x}_{T-1}, \dots, \mathbf{x}_0)$ is generated by repeatedly sampling $\mathbf{x}_{t-1} \sim \pi_\theta(\cdot | \mathbf{x}_t, c)$ for $t = T, \dots, 1$.

4.2 Sparse Incremental Timestep Update

We use the DDIM formulation and first consider the role of structural regularization via selected timestep updates [2, 24]. We denote the list of updated timesteps as $\mathcal{S} = t_i$. Below we provide actionable theoretical results followed by specific design principles implied thereof. The proofs of Propositions 1, 2 and Remark 2 are provided in the supplementary.

Proposition 1 (Generative Reach). *Let $R(\mathbf{x}_0, c)$ be a sparse terminal reward and $\delta\epsilon_t$ be the parameter update induced by the RL objective at timestep t . For a fixed budget of $N = |\mathcal{S}|$ optimized timesteps, a uniform sampling strategy $\mathcal{S}_{\text{uni}} \sim \mathcal{U}(1, T)$ yields a strictly greater total perturbation magnitude $\|\delta\mathbf{x}_0\|$ than a late-stage strategy $\mathcal{S}_{\text{late}} \sim \mathcal{U}(1, N)$.*

Early timestep update. The above result directly indicates that early timesteps should be updated in the RL finetuning of diffusion models, contrary to the conclusions of previous works [2, 24].

Proposition 2 (Volume Preservation via Temporal Sparsity). *Let $\mathcal{J}_{T \rightarrow 0}$ be the total Jacobian of a T -step deterministic reverse diffusion process. For a*

fixed RL optimization budget, a sparse update strategy over a subset $\mathcal{S}$ (where $|\mathcal{S}| = N \ll T$) preserves the probability volume of the prior distribution strictly better than a full update strategy ($\mathcal{S} = \{1, \dots, T\}$), as measured by the log-determinant of the global Jacobian.

Remark 1 (Capacity and Over-parametrization). Proposition 2 can be interpreted through the lens of model capacity. Since distinct diffusion timesteps $\mathcal{S}$ specialize in non-overlapping spectral components of the image manifold [11], a full update strategy ($\mathcal{S} = \{1, \dots, T\}$) effectively increases the degrees of freedom available to the RL optimizer. By optimizing this over-parameterized "temporal ensemble," the model more easily finds degenerate solutions that satisfy the terminal reward R at the cost of global prior collapse.

Sparse timestep update. The above result clearly indicates that selecting sparse timesteps for update provides strong regularization in policy optimization of the diffusion models. In fact, this also provides a tangible explanation for the improved performance of [24] when using only late timesteps with additional components of the approach.

Remark 2 (Synthesis: Temporal Curriculum of Generative Reach). The relationship between generative reach (Prop. 1) and volume preservation (Prop. 2) implies an optimal curriculum for reward finetuning: an incremental expansion of the active timestep subset $\mathcal{S}$. By initially restricting optimization to a sparse set of high-leverage early timesteps ($t \sim T$), the model achieves global mode discovery with minimal cumulative contraction, only increasing $|\mathcal{S}|$ to refine local manifold details once the distribution has stabilized. Fig.3(B) visually illustrates the resulting sparse incremental timestep updates.

4.3 Feynman-Kac Path Guidance

An important drawback of any policy optimization strategy is the sample inefficiency due to terminal rewards obtained only at $\mathbf{x}_0$. However, the Feynman-Kac theory [9,28] provides a way out by considering each path's prospect at any given timestep. While previous works have used discrete-time DDIM with FK guidance for inference-time sample improvement [50,52], here, we devise an approach for training.

Feynman-Kac (FK) view and intermediate potentials. Optimizing under the FK guidance requires defining a measure that evaluates a path's reward satisfaction, termed as the potential. Following FK steering [51,52], we introduce a sequence of nonnegative *potentials* $\{G_t\}_{t=1}^T$ that provide intermediate weighting of partial trajectories. Concretely, given a trajectory τ , we associate the unnormalized weight

$$w(\tau) = \prod_{t=1}^T G_t(\mathbf{x}_t, c), \quad (6)$$

and define an FK-reweighted path distribution

$$\tilde{p}_\theta(\tau) \propto p_\theta(\tau) \prod_{t=1}^T G_t(\mathbf{x}_t, c), \quad (7)$$

where $p_\theta(\tau)$ is the trajectory distribution induced by π_θ and the diffusion prior on $\mathbf{x}_T$.

Rather than re-evaluating a denoised prediction at every step, we use FK-style potentials to *allocate compute across a branching sampling tree*. Starting from an initial state $\mathbf{x}_T$, we generate a set of partial trajectories by branching the reverse diffusion transitions. We then complete each branch to obtain terminal samples $\mathbf{x}_0$ and evaluate the terminal reward $R(\mathbf{x}_0, c)$. The resulting terminal rewards are used to assign a scalar potential to each branch (and, by aggregation, to its intermediate nodes), which allows us to prune branches that are unlikely to yield high reward. Fig. 3 clearly illustrates the path termination and branching for improving alignment and diversity respectively. We provide the detailed algorithm in the supplementary material.

4.4 Rare Events and Rarity

In order to correctly quantify desired shifts in distribution, we formalize the concept of rarity. Given a condition c , let $p_\phi(\mathbf{x}_0 | c)$ denote the pretrained model’s distribution induced by the reverse diffusion sampler. We define the rare-event set as:

$$\mathcal{A}_c(\eta) = \{\mathbf{x}_0 : R(\mathbf{x}_0, c) \geq \eta\}. \quad (8)$$

An event is rare under p_ϕ when $\mathbb{P}_{\mathbf{x}_0 \sim p_\phi(\cdot | c)}(\mathbf{x}_0 \in \mathcal{A}_c(\eta))$ is small [9, 16, 51].

We refer to a condition c as a *rare condition* when it triggers rare events under the sampler, *i.e.*, when the probability mass over high-reward outcomes is small. We define the *rarity* of a condition as the rare-event probability,

$$\rho(c; \eta) = \mathbb{P}_{\mathbf{x}_0 \sim p_\phi(\cdot | c)}(\mathbf{x}_0 \in \mathcal{A}_c(\eta)). \quad (9)$$

Smaller $\rho(c; \eta)$ indicates higher rarity (harder conditions). For a new model with distribution $p_\theta(\mathbf{x}_0 | c)$ induced by the model π_θ , we define rarity and rare conditions with the same event set $\mathcal{A}_c(\eta)$ and threshold η ; only the underlying sampling distribution changes from p_ϕ to p_θ .

5 Experiments

5.1 Experimental Setup

Image Generation. For image generation, we use Stable Diffusion (SD) v1.4 [45] and fine-tune only the UNet via Low-Rank Adaptation (LoRA) [21] for parameter-efficient adaptation, following DDPO [4]. We evaluate prompts from three image-domain categories: compositional generation (animals & actions), attribute binding (colors & fruits), and spatial relations similar to B2DiffuRL [24]. Representative prompts include “a whale riding a bike”, “brown banana”, and “a cup on

tshirt”. We use the DDIM sampler [55] with settings consistent with prior RL fine-tuning work [4, 24]; unless otherwise stated, we use $T = 20$ denoising steps and classifier-free guidance scale $w = 5.0$. The image reward is CLIPScore, *i.e.*, the similarity between text and image embeddings measured by CLIP [43]. The full prompt list is reported in the supplementary material. We employ an incremental training curriculum with four stages using $\mathcal{T}_{\text{sch}} = \{5, 10, 15, 20\}$ progressively increasing the updated timesteps. We train each stage for 10 iterations. For FK-steered sampling, we use $k = 4$ particles per prompt and resample every 5 denoising steps (*i.e.*, $\mathcal{T}_{\text{res}}$ with period 5), with FK scale $\lambda_{\text{FK}} = 2.0$.

Vanishing Point (VP) Correction. VP experiments use the same image-generation pipeline and sampler settings as the general synthesis task to ensure controlled comparisons. We evaluate prompts focused on long-range geometric consistency (e.g., “railway tracks” or “tunnel scenes with clear perspective cues”). We define the reward as the algebraic consensus of the straight lines converging to distinct vanishing points. We call this reward *VP Geometry*. We detect line segments using Line Segment Detection (LSD) [31] and apply a custom line grouping algorithm. Additional algorithmic details are in the supplementary material.

3D Indoor Scene Synthesis. We adopt the continuous-domain variant of MiDiffusion [22] as the base diffusion model and use the 3D-FRONT dataset [13]. Following prior 3D scene-generation work [22, 39, 57], each scene is represented as an unordered set of objects parameterized by semantic class, 3D position, size, and rotation. We train for 2 different rewards separately. First, we define a custom spatial reward for the prompt “a scene with a TV stand facing a bed,” which evaluates both object position and orientation. Second, we use a collision reward [40] that assigns +1 to valid scenes and otherwise applies a penalty equal to the total axis-aligned penetration depth across overlapping ground objects. Additional implementation details are provided in the supplementary material.

Methods Compared. We compare four different models including the pretrained base reference model. We then compare two RL baselines: DDPO [4], and its backward progressive extension B²-DiffuRL [24]. Finally, we evaluate our method, iADD. All methods start from the same pretrained base model.

5.2 Evaluation Metrics

Alignment–Diversity. We quantify alignment using the task-specific reward models defined in Sec. 5.1. We measure image diversity using the Inception Score (IS), following [24]. We additionally report area-under-curve (AUC) metrics to summarize the trade-off between these objectives across operating points. Higher AUC indicates a better overall balance between alignment and diversity.

Rarity (Rare-Event Probability). To quantify performance on rare or difficult conditions, we first fix the rare-condition set $\mathcal{C}_{\text{rare}}$ from the frozen pretrained model and keep this split identical for all methods. Following Sec. 4.4, rarity for

each condition is defined by the model-specific rare-event probability under the same event set $\mathcal{A}_c(\eta)$ and threshold η .

We only report the aggregated rarity score $\text{Rarity}(m) = \frac{1}{|\mathcal{C}_{\text{rare}}|} \sum_{c \in \mathcal{C}_{\text{rare}}} \hat{\rho}_m(c; \eta)$, where $\hat{\rho}_m(c; \eta)$ is the empirical rare-event probability for method m (estimated from K samples per condition; $K = 24$ in image experiments), consistent with Sec. 4.4. Following the Q3 rule, η is set to the 75th-percentile reward of the frozen pretrained reference distribution for each task, which yields $\eta = 0.34$ for image generation and $\eta = 0.64$ for vanishing-point correction. Higher $\text{Rarity}(m)$ indicates higher rare-event success probability on the fixed rare-condition set.

3D-Specific Metrics. We report Collision ($\downarrow$), using the same penetration-based formulation as the collision reward defined in Sec. 5.1, and Class Label KL Divergence (CKL, $\downarrow$). The CKL reference distribution is computed from 3D-FRONT [13] test scenes that satisfy the spatial reward constraint. To ensure a fair comparison, we evaluate all methods at a common convergence point where they reach the same mean reward threshold during training.

Evaluation Protocol. We evaluate models at regular intervals across 45 base prompts, generating 1,080 images per checkpoint using deterministic seeds to ensure robust measurement. To guarantee a fair comparison, we report each method at its optimal operating point—defined as the checkpoint closest to the ideal (1,1) on the Reward-Inception Score Pareto frontier.

5.3 Qualitative Assessment

We qualitatively compare our method against SD, DDPO, and B²-DiffuRL across all three tasks. Unless otherwise noted, all visualizations are generated from the best operating point of each model after training.

Figure 4 shows results on common and rare prompts. On common prompts, DDPO and B²-DiffuRL improve alignment, but our method produces sharper details and more realistic textures while preserving prompt fidelity. On rare prompts, DDPO and B²-DiffuRL either fail to satisfy the prompt or produce cartoonish outputs, whereas our method maintains both semantic alignment and visual quality. These observations are consistent with the quantitative reward-diversity trends reported in Table. 3. We report more common and rare prompts in supplementary.

Figure 5b shows that our method yields line structures that converge more consistently to a single vanishing point, while DDPO and B²-DiffuRL often produce weaker or inconsistent convergence. In particular, the first-row DDPO example loses texture fidelity while still failing to enforce clean perspective geometry. Our method preserves texture and local details while satisfying the perspective constraint. This can be attributed to our incremental training schedule with FK-based reweighting and FK branching, which propagates geometric rewards across timesteps without collapsing visual quality.

Figure 5a presents qualitative 3D indoor scene results. For 3D, we evaluate a prompt with explicit geometric constraints ("a scene with TV stand facing

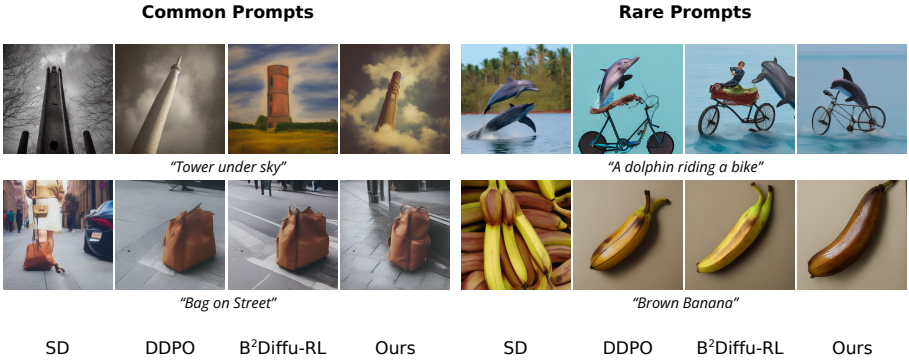


Fig. 4: Qualitative comparison on common and rare prompts. Our method better preserves prompt alignment while maintaining realistic visual quality, with larger gains on rare prompts.

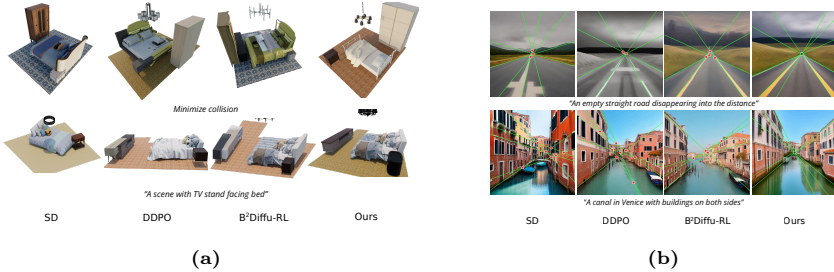


Fig. 5: Qualitative results. (a) 3D scene synthesis. Top: collision reward. Bottom: spatial reward (“TV stand facing bed”). DDPO and B²-DiffuRL satisfy constraints but introduce collisions; our method avoids them. **(b) Vanishing-point correction.** Red: off-vanishing intersections; green: vanishing point. DDPO and B²-DiffuRL shows weaker convergence and desaturation; our method preserves quality while enforcing perspective. (Better viewed zoomed.)

bed”) (see more prompts in supplementary). Even when optimizing rewards not directly tied to collision minimization, our method better preserves non-colliding layouts than DDPO and B²-DiffuRL, as visible in Figure 5a in the 2nd row and consistent with the observation from Figure 6a which reports the corresponding reward–collision trade-off. This shows reward hacking in iADD is lower than other baselines.

Human Evaluation. We evaluate perceptual quality and prompt fidelity with a human-preference study over image Templates 1–3. Following the operating-point selection in Sec. 5.2, we sample one image per method for each of 30 prompts and ask 40 independent raters to choose the best output in a blind comparison among the four methods (ours, B²-DiffuRL, DDPO, and the pre-trained baseline), yielding 4,800 total judgments. As shown in Figure 6c, our method is preferred more frequently than DDPO and B²-DiffuRL.

Table 3: Performance comparison across metrics and modalities. Evaluated at their best operating points, our method (iADD) consistently achieves the best alignment-diversity trade-off across all three tasks compared to DDPO and B^2 -DiffuRL. Notably, iADD maintains higher diversity (better IS and lower collision rates) at comparable or higher reward levels, while also demonstrating state-of-the-art robustness on rare prompts (Rarity). Arrows indicate the desired direction.

Method	Image Generation				3D Indoor Scene		Vanishing Point Correction		
	Reward $\uparrow$	IS $\uparrow$	Rarity $\uparrow$	AUC $\uparrow$	Collision $\downarrow$	CKL $\downarrow$	Reward $\uparrow$	IS $\uparrow$	Rarity $\uparrow$
Pretrained (SD)	0.3272	1.3514	-	-	53.87	30.48%	0.6105	1.2735	-
DDPO	0.3417	1.2786	24.81%	53.8%	63.39%	7.726%	0.6815	1.2600	21.33%
B^2 -DiffuRL	0.3452	1.2728	24.22%	58.7%	64.06%	6.278%	0.6486	1.2550	31.11%
Ours	0.3553	1.3241	66.85%	77.7%	59.20%	6.13%	0.7763	1.2881	95.11%

5.4 Quantitative Results

Alignment–Diversity Trade-off. Figure 1a plots Inception Score (IS) versus CLIP-Score across training for each method. As expected, pushing alignment tends to reduce diversity/quality; however, DDPO and B^2 -DiffuRL show substantially larger IS degradation at comparable reward levels. In contrast, our method maintains a better reward–IS tradeoff. This is metrically reported in Table 3.

Rarity. We report rarity in Table 3 for both image generation and vanishing-point correction. iADD achieves the strongest rarity performance among all methods. Figure 1b shows significant improvement of iADD on rare prompts.

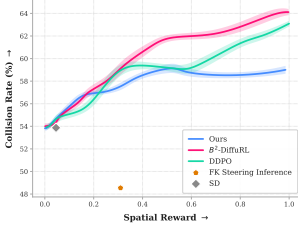
3D Indoor Scene Synthesis. Table 3 shows that, under the same reward target ("a scene with TV stand facing bed") and stopping criterion, DDPO and B^2 -DiffuRL produce higher collision rates than our method. Figure 6a further illustrates this trend across training: competing methods obtain reward gains at the cost of increased collisions, whereas our method maintains lower collision while preserving competitive reward.

5.5 Ablation Studies

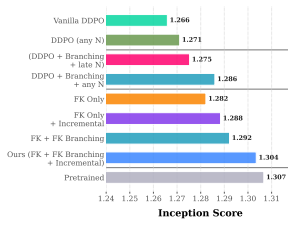
We conduct controlled ablations to isolate important components for better alignment–diversity–rarity. We present our findings in Table 1.

Timestep Selection Ablation. We first compare standard DDPO with 20-step updates *Vanilla DDPO* against a sparse-update variant that updates any five timesteps *DDPO (any N)*, $N = 5$ matched at comparable reward levels. As shown in Fig. 6b, *DDPO (any N)* achieves higher diversity, supporting Prop. 2.

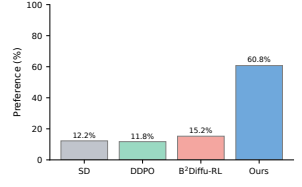
Next, we compare timestep schedules in branching: *DDPO+Branching+late N* versus *DDPO+Branching+any N*. Updating any N timesteps consistently yields higher IS than late- N updates (Fig. 6b), suggesting that the late-timestep strategy used in B^2 -DiffuRL [24] is suboptimal, as predicted by Prop. 1.



(a) **Spatial reward vs. collision.** Training with spatial reward only (“TV stand facing bed”). Our method achieves fewer collisions at matched reward levels.



(b) **Diversity ablation (IS).** Incremental scheduling and FK branching improve diversity over DDPO variants, with our method achieving the best score.



(c) **Human preferences.** Pair-wise ratings from 40 raters on Templates 1–3. Our method is preferred over DDPO and B²-DiffuRL.

Ablations on Incremental Strategy. Motivated by the gains from any- N updates, we introduce an incremental (curriculum-style) timestep schedule. We compare *FK only* against *FK + Incremental*. Figure 1b shows improved alignment on both common and rare prompts, while Fig. 6b shows improved diversity. We attribute this behavior to stronger reward propagation into earlier denoising steps, which mitigates sparse-reward optimization and encourages broader exploration.

Ablations on FK and FK Branching. We evaluate the effect of Feynman–Kac reweighting (FK) and FK Branching on alignment and diversity. Figure 1b shows that both FK-based variants improve alignment on common and rare prompts compared with DDPO, when evaluated at a matched diversity target (IS $\approx$ 1.28). The CLIPScore gap between *FK only* and *FK + Branching* is modest; however, *FK + Branching* consistently attains higher diversity (Fig. 6b). This is consistent with the intuition that branching replicates high-potential particles and improves mode coverage during training [24]. Moreover, the CLIPScore increase from *DDPO+Incremental* to *Ours (FK Only)* in Fig. 1b further indicates that Feynman–Kac reweighting improves *Rarity*.

6 Conclusions

Our work tackles the problem of jointly tackling alignment-diversity-rarity improvement in diffusion models via reinforcement learning. We showed with theoretical and empirical evidence that careful considerations should be made in timestep selection during reward optimization to avoid quality degradation. Thus we proposed an incremental sparse timestep update strategy to improve the alignment-diversity tradeoff. Furthermore, we provided a discrete implementation of Feynman–Kac path guidance through path pruning and branching thus jointly improving all three metrics. Finally, we verified our approach iADD with ablations as well as provided comparisons against the state-of-the-art in three different tasks validating its task-agnostic usefulness.

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